MIGHTTEAM FPPW2
6INTELSY March 20, 2026

PPE DETECTION SYSTEM
DOCUMENTATION WEEK 2

Data Acquired, Splits, and EDA Notebook

- Dataset: The dataset used is the Roboflow PPE dataset, with its license verified.
- Splits: The train, validation, and test splits were confirmed using data.yaml, and all labels were verified.
- Data quality: There are no missing or corrupt labels or images, class balance was reviewed, and varied image resolutions are handled through resizing.
- EDA: The class distribution and bounding box sizes were analyzed, showing a dominance of small to medium objects, and no data leakage issues were found.
- Notebook: The EDA was conducted in notebooks/eda.ipynb, with outputs saved in results/eda/.
- Evidence: The data/ folder structure and sample images have been documented as supporting evidence.

Baselines Trained

- Classical baseline: The classical baseline was intentionally skipped to prioritize a deep learning approach better suited for object detection tasks.
- DL baseline: A YOLOv8n model was trained for 10 epochs as the deep learning baseline.
- Metrics recorded: Performance metrics including mAP@0.5, mAP@0.5:0.95, precision, and recall were successfully logged.
- What worked / failed: The model achieved fast training and reasonable detection performance, but accuracy was limited due to the lightweight architecture and short training duration.
- Links: The training command is documented in the README, and logs are saved in runs/detect/ (Ultralytics outputs).

CNN Experiment

- CNN experiment running (first results): A YOLOv8s model is being trained for ~40 epochs with standard 640px input size and default augmentations, achieving initial mAP@0.5, mAP@0.5:0.95, and latency measurements.
- Issues: Minor class imbalance and slightly lower recall on small PPE items were observed.
- Next tweaks: Planned improvements include learning rate tuning, augmentation adjustments, and exploring anchor-free or improved loss variants.
- NLP component scaffolded/prototyped: The NLP component (scripts/nlp_component_stub.py) has finalized alert text templates, sample outputs recorded, and is ready for integration with detection results to generate real-time alerts.

RL Component (Stub + Reward Design)

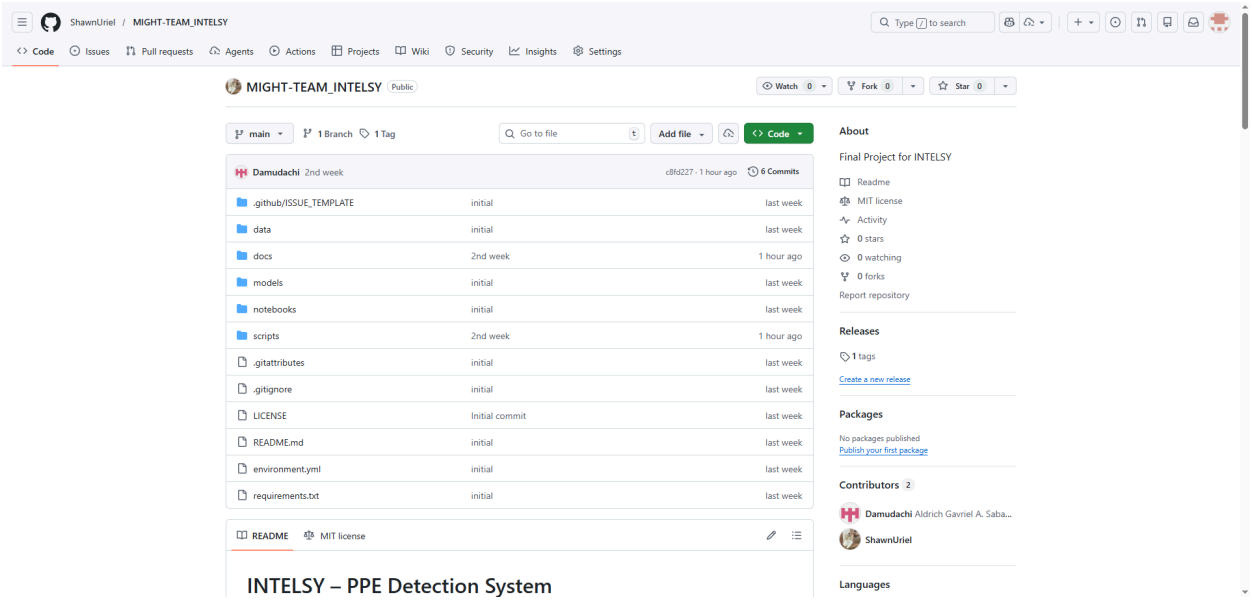
- RL agent stubbed with reward design; early learning curves (even if noisy): An RL agent stub has been implemented in scripts/rl_stub.py with rewards of +1 for correctly flagged PPE and penalties for misses or false alarms; reward shaping is defined for balanced learning.
- Status: The stub has been executed, but early learning curves have not yet been produced.
- Next: The RL agent will be integrated with the detection pipeline, and training loop experiments will begin.

Ethics & Model Card

- Draft Model Card sections started; Ethics statement updated: The model card has been drafted in docs/model_card.md, including architecture, dataset info (source, splits, license), hyperparameters, metrics, failure modes, and ethical considerations, while the ethics statement was updated to note dataset license, no sensitive data misuse, deployment warnings, and bias risks.

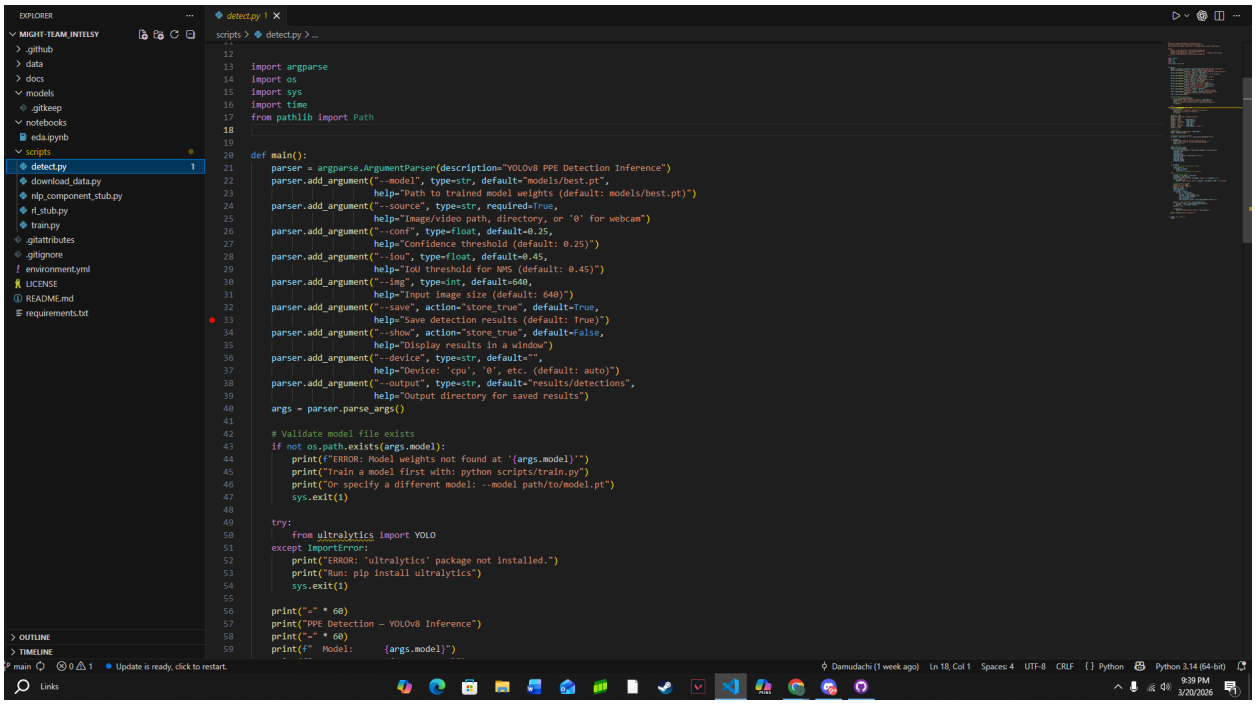
Deliverables Checklist

Github updated repo:



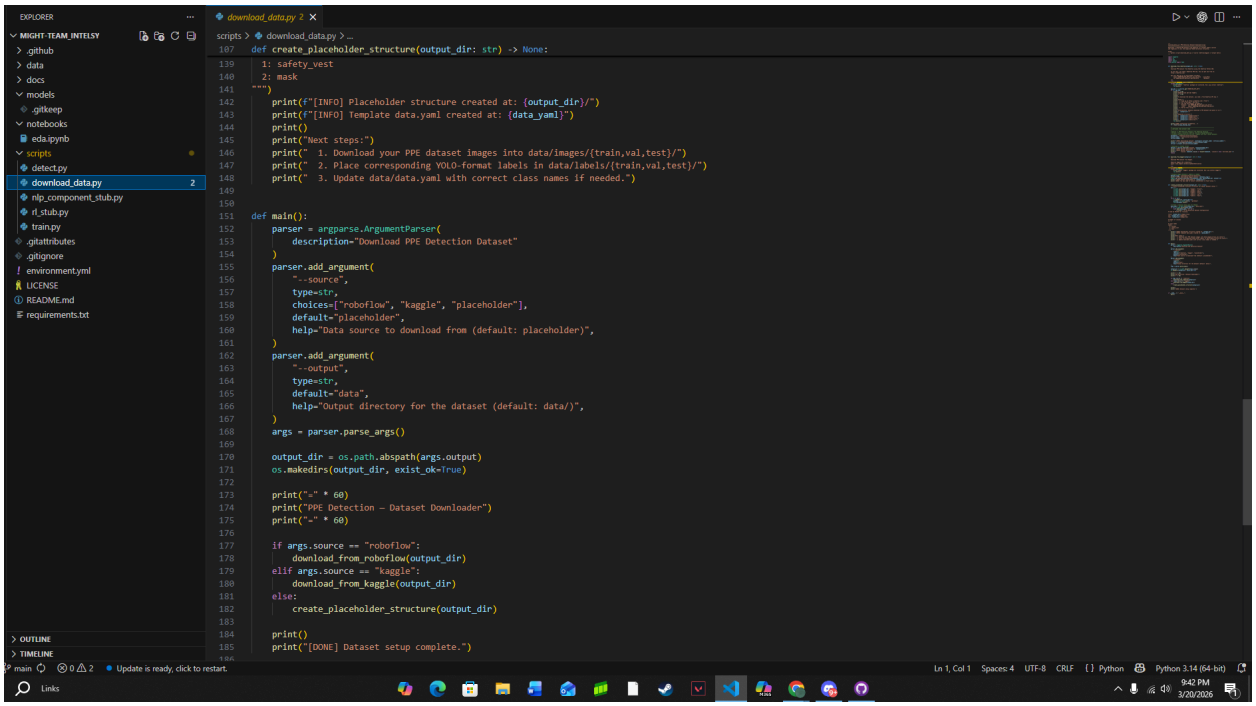
Code Screeshots:

detect.py

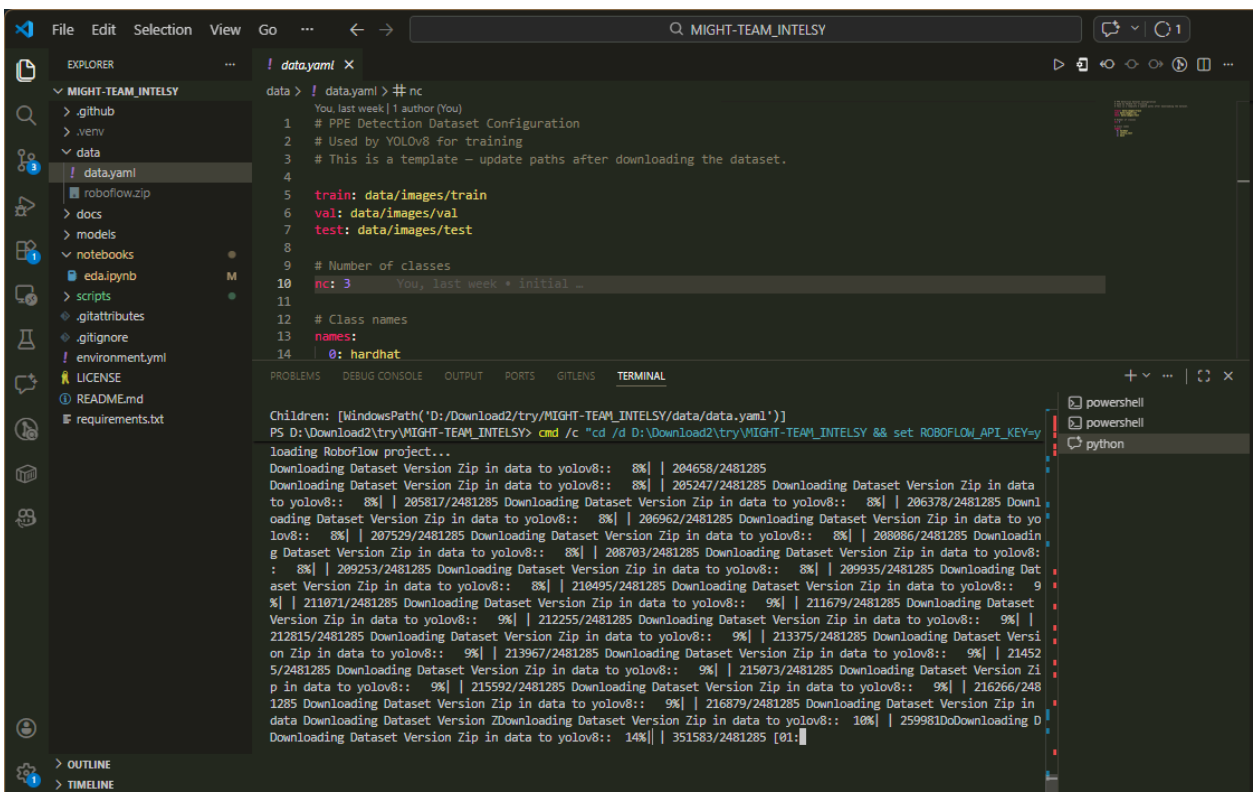


```python
import argparse
import os
import sys
import time
from pathlib import Path


def main():
    parser = argparse.ArgumentParser(description="YOLOv8 PPE Detection Inference")
    parser.add_argument("--model", type=str, default="models/best.pt",
                        help="Path to trained model weights (default: models/best.pt)")
    parser.add_argument("--source", type=str, required=True,
                        help="Image/video path, directory, or '0' for webcam")
    parser.add_argument("--conf", type=float, default=0.25,
                        help="Confidence threshold (default: 0.25)")
    parser.add_argument("--iou", type=float, default=0.45,
                        help="IoU threshold for NMS (default: 0.45)")
    parser.add_argument("--img", type=int, default=640,
                        help="Input image size (default: 640)")
    parser.add_argument("--save", action="store_true", default=True,
                        help="Save detection results (default: True)")
    parser.add_argument("--show", action="store_true", default=False,
                        help="Display results in a window")
    parser.add_argument("--device", type=str, default="",
                        help="Device: 'cpu', '0', etc. (default: auto)")
    parser.add_argument("--output", type=str, default="results/detections",
                        help="Output directory for saved results")
    args = parser.parse_args()

    # Validate model file exists
    if not os.path.exists(args.model):
        print(f"ERROR: Model weights not found at '{args.model}'")
        print("Train a model first with: python scripts/train.py")
        print("Or specify a different model: --model path/to/model.pt")
        sys.exit(1)

    try:
        from ultralytics import YOLO
    except ImportError:
        print("ERROR: 'ultralytics' package not installed.")
        print("Run: pip install ultralytics")
        sys.exit(1)

    print("=" * 60)
    print("PPE Detection — YOLOv8 Inference")
    print("=" * 60)
    print(f"  Model:     {args.model}")
```

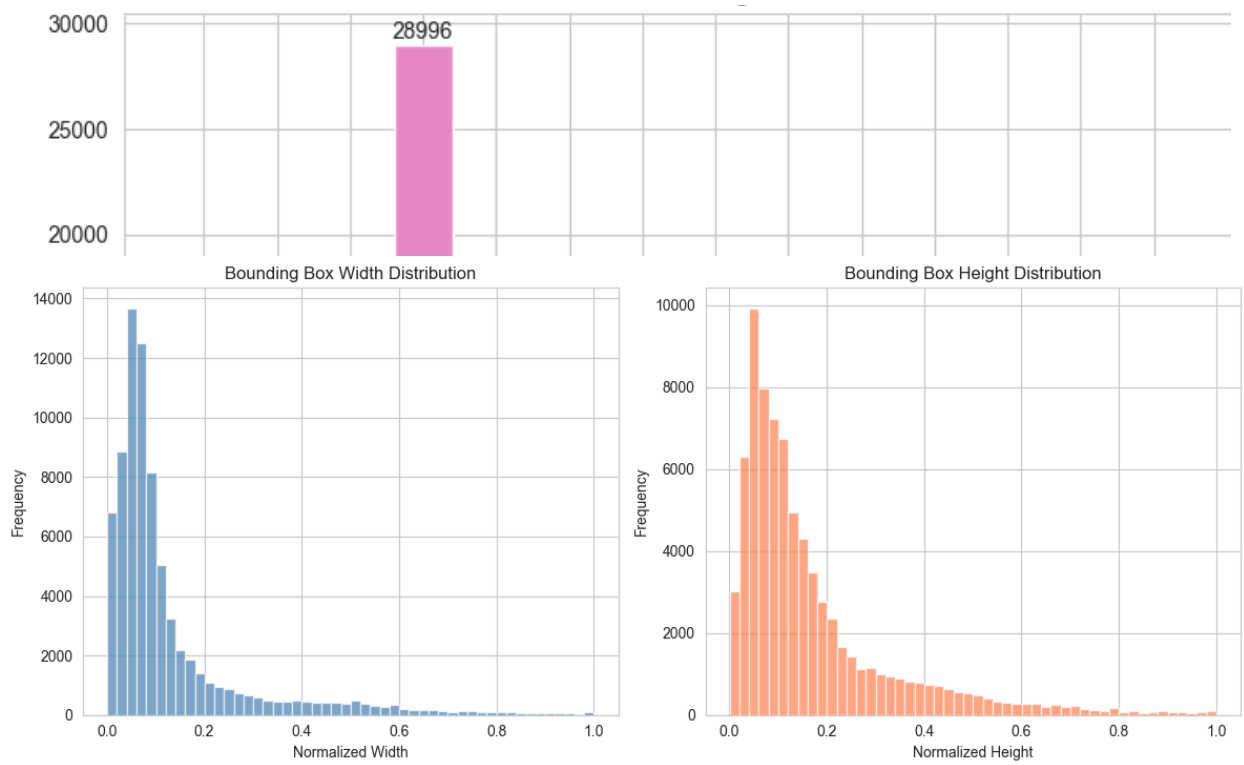
download_data.py



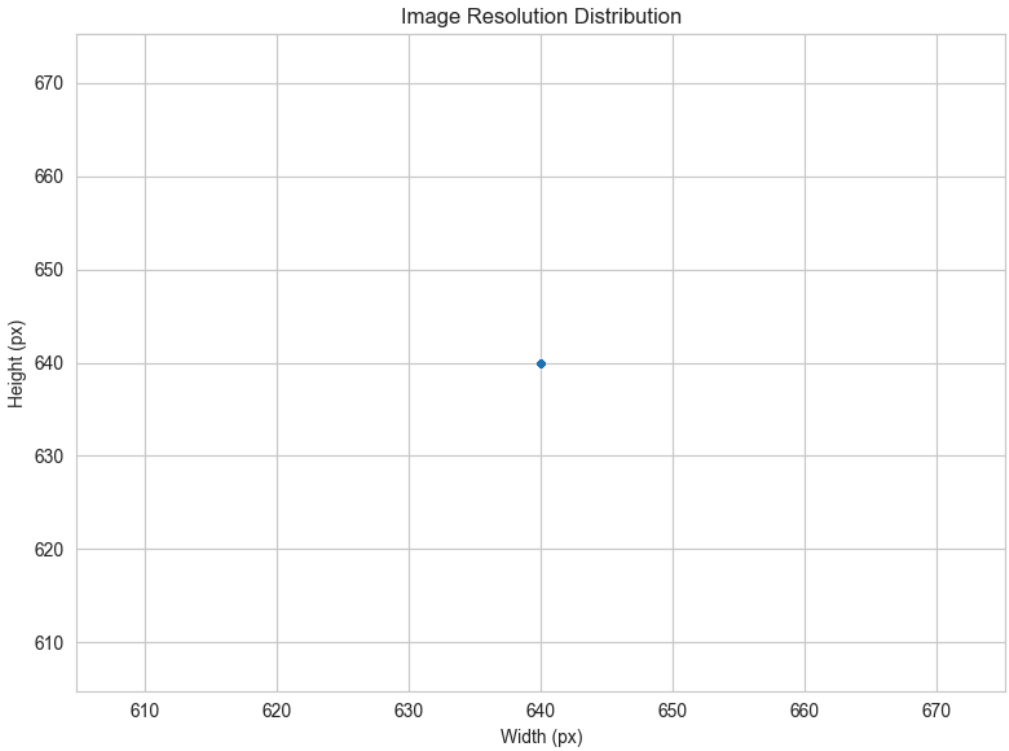
dataset download:



Class Distribution in Training + Validation Set



Sample training images:



Width — min: 640, max: 640, mean: 640
Height — min: 640, max: 640, mean: 640